Day 2 Reference Guide: IAM and AWS Budgets

Cloud Unlock · Sunday, 11 October 2026 · Services, key ideas and official AWS links

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Live class: Sunday 11 Oct 2026, 7:00-9:00 PM IST
Exam focus: Domain 2 (Security and Compliance), Domain 4 (Billing)
Exam tasks: 2.3 · 4.2

Regions, Availability Zones and edge locations were taught on Day 1. The AWS CLI and shared responsibility come on Day 3.

What You Will Know

- You have a budget alert and know that budgets alert, they do not stop spending.
- Your admin access comes from `AdminGroup`, your IAM user has MFA, and you work as `Cloud-Unlock` (not root).
- You can explain IAM users, groups, policies and roles in one line each.
- You can read a simple IAM policy: `Effect`, `Action`, `Resource`.
- You practised least privilege with three users (S3, EC2, billing): what each can and cannot do.
- You know when to use a role (temporary access; an app on EC2 uses a role, not access keys) and the three policy types.

Day 2 - IAM And AWS Budgets: Key Ideas

Goal: understand how AWS decides who can do what.

Exam tasks: 2.3 (AWS access management), 4.2 (billing tools)

Remember this:

> *Identity + Permissions = Access*

The Story

Think of your **college ERP**. A student sees only their own marks. Faculty can enter marks. The admin manages everything. Same app, different access. That is what IAM does for your AWS account.

IAM In Simple Words

IAM word	One line	College ERP example
User	One person with a login	You, with your student login
Group	A team of users with the same access	All faculty
Policy	JSON rules that allow or deny actions	"Faculty can edit marks"
Role	Temporary access for a task, no password	A visitor pass that expires
Root user	The sign-up email, full access	The college owner's master key

Login access does not mean full access. **Policies decide what a user can do.**

- **Authentication** = proving who you are (sign in).
- **Authorization** = what you are allowed to do (policies).

IAM Users

- A named identity for one person (or an application).
- Signs in to the Console with a password, and can have MFA.
- Example from Day 1: `Cloud-Unlock` is your daily admin user.

IAM Groups

- A collection of users with common permissions.
- Attach the policy to the **group**, then add users to the group.
- A group cannot contain another group.

Examples we use today:

Group	Policy	User	Can do	Cannot do
`AdminGroup`	`AdministratorAccess`	`Cloud-Unlock`	Everything (daily admin)	Root-only tasks
`S3ReadOnlyGroup`	`AmazonS3ReadOnlyAccess`	`Cloud-Unlock-S3`	View buckets and files	Create or delete
`EC2ReadOnlyGroup`	`AmazonEC2ReadOnlyAccess`	`Cloud-Unlock-EC2`	View EC2	Launch, stop, terminate
`BillingViewGroup`	`AWSBillingReadOnlyAccess`	`Cloud-Unlock-Billing`	View bills and costs	Manage S3 or EC2

Exam pointer: "manage permissions for many users" = **IAM groups**.

IAM Policies

An IAM policy is a JSON document. It has three main words:

```
{
  "Version": "2012-10-17",
  "Statement": [
    { "Effect": "Allow", "Action": ["s3:Get*", "s3:List*"], "Resource": "*"
}
  ]
}
```

Word	Meaning
`Effect`	`Allow` or `Deny`
`Action`	What can be done, for example `s3:ListBucket`
`Resource`	On which resource, `*` means all

Types:

- **AWS managed policy:** created and maintained by AWS, for example `AmazonS3ReadOnlyAccess`.
- **Customer managed policy:** created by you and reusable.
- **Inline policy:** attached directly to one user, group or role, and deleted with it.

Rule: an explicit **Deny** always wins over an Allow. If nothing allows an action, it is denied by default.

IAM Roles

- A role gives **temporary** credentials. No password, no long-term keys.
- Used when:

    - an AWS service needs to call another AWS service (for example, an EC2 server reading S3);
    - a person switches into a role for a task.

Exam pointer: "an application on EC2 needs to access S3" = **IAM role**, not access keys in code.

Least Privilege

Give only the permissions needed for the task.

- Needs to view S3 only → `AmazonS3ReadOnlyAccess`, not full access.
- Needs to view EC2 only → `AmazonEC2ReadOnlyAccess`, not `AdministratorAccess`.
- Finance needs to see the bill → `AWSBillingReadOnlyAccess`, not access to servers.
- A Myntra support agent can see an order but cannot delete the catalogue.

Keeping Access Safe

- **MFA** on the root user and on every IAM user.
- **Access keys** are for the CLI or code. Never share them, never put them on GitHub. Never create them for root.
- **Password policy:** set a minimum length and require numbers and symbols.
- Do daily work as an IAM user. Use root only for root-only tasks.

AWS Budgets And The Cost Tools

Service	What it does	When you use it
AWS Budgets	Sends an alert when cost or usage crosses your limit	While you run things
AWS Cost Explorer	Shows past spend and a forecast	After you spend
AWS Pricing Calculator	Estimates cost before you build	Before you build

Exam pointer: a budget **alerts**; it does **not** stop spending.

Official Links

Topic	Link
What is IAM	https://docs.aws.amazon.com/IAM/latest/UserGuide/introduction.html
IAM best practices	https://docs.aws.amazon.com/IAM/latest/UserGuide/best-practices.html
IAM groups	https://docs.aws.amazon.com/IAM/latest/UserGuide/id_groups_create.html
Policies and permissions	https://docs.aws.amazon.com/IAM/latest/UserGuide/access_policies.html
IAM roles	https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles.html
AWS Budgets	https://docs.aws.amazon.com/cost-management/latest/userguide/budgets-managing-costs.html

Topic	Link
IAM roles for Amazon EC2	https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/iam-roles-for-amazon-ec2.html
Billing access for IAM users	https://docs.aws.amazon.com/awsaccountbilling/latest/aboutv2/control-access-billing.html

More Official Links

Topic	Link
Create an IAM user	https://docs.aws.amazon.com/IAM/latest/UserGuide/id_users_create.html
Create IAM groups	https://docs.aws.amazon.com/IAM/latest/UserGuide/id_groups_create.html
Billing access for IAM users	https://docs.aws.amazon.com/awsaccountbilling/latest/aboutv2/control-access-billing.html
Switch to a role in the console	https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles_use_switch-role-console.html
Password policy	https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_passwords_account-policy.html

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